

Nomogram for visual acuity outcome after iodine-125 plaque radiotherapy and prophylactic intravitreal bevacizumab for uveal melanoma in 1131 patients

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► Additional material is published online only. To view please visit the journal online (<http://dx.doi.org/10.1136/bjophthalmol-2019-314686>).

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Received 31 May 2019

Revised 16 July 2019

Accepted 21 July 2019

Published Online First

13 August 2019

ABSTRACT

Aims To develop a nomogram for prediction of visual acuity outcome following plaque radiotherapy for uveal melanoma.

Methods Retrospective review of uveal melanoma treated with plaque radiotherapy and prophylactic intravitreal bevacizumab injections at 4-month intervals for 2 years duration. Two nomograms for poor visual acuity outcome (Snellen <20/200) were developed based on (1) Clinical risk factors. (2) Or clinical and treatment risk factors.

Results There were 1131 included cases. The most important clinical risk factors (points for nomogram) for poor visual acuity outcome included subretinal fluid involving four quadrants (100), tumour thickness >4 mm (69), presenting visual acuity ≤20/30 (65), non-Caucasian race (58), tumour shape mushroom, bilobed, or multilobulated (57), and insulin-dependent diabetes (54). Risk of poor visual acuity at 2 years and 4 years increased from 11% and 24% with 40 points to 97% and >99% with 304 points. A second analysis was performed using both clinical and treatment risk factors. The most important factors included presenting visual acuity ≤20/30 (100), tumour largest basal diameter >11 mm (80), radiation dose rate to tumour base ≥164 cGy/hour (78), tumour thickness >4 mm (76), insulin-dependent diabetes (75) and abnormal foveolar status by optical coherence tomography at presentation (72). Risk of poor visual acuity at 2 years and 4 years increased from 6% and 14% with 56 points to 88% and 99% with 496 points.

Conclusions A nomogram using clinical or treatment risk factors can predict visual acuity outcome following plaque radiotherapy and prophylactic intravitreal bevacizumab for uveal melanoma and is available online at <https://fighteyecancer.com/nomograms/>.

INTRODUCTION

Plaque radiotherapy is an effective treatment modality for uveal melanoma, but radiation side effects can lead to poor visual acuity outcome in up to 68% of cases after 10 years.^{1–3} More recently, prophylactic intravitreal bevacizumab has been used to prevent postradiation macular oedema and protect visual acuity following plaque radiotherapy for uveal melanoma.³ Compared with control patients who were managed without bevacizumab, Shah *et al* found reduced risk of poor visual acuity outcome at 2 years follow-up in patients treated with prophylactic intravitreal bevacizumab at the

time of plaque removal and every 4 months for a duration of 2 years.³

While intravitreal antivascular endothelial growth factor (anti-VEGF) therapy can mitigate some radiation side effects, including macular oedema, multiple factors contribute to poor visual acuity outcome in these patients.¹ Khan *et al* developed a nomogram to predict poor visual acuity outcome worse than 20/200 following plaque radiotherapy in 189 eyes with uveal melanoma and found significant predictors of increased age, tumour thickness and radiation dose to the macula.⁴ The nomogram included those treated with either ruthenium-106 (Ru-106) or iodine-125 (I-125) and was developed prior to the implementation of prophylactic anti-VEGF.⁴ Thus, further investigation to develop a more relevant nomogram most predictive of poor visual acuity outcome in the current era of prophylactic anti-VEGF injections using a larger series of patients is warranted. Herein, we present two nomograms for visual acuity outcome based on clinical and treatment risk factors in a large series of patients treated with I-125 plaque radiotherapy and prophylactic intravitreal bevacizumab.

METHODS

Medical records were reviewed to identify all patients diagnosed with uveal melanoma at the Ocular Oncology Service, Wills Eye Hospital, Thomas Jefferson University, from 7 July 2000 to 1 January 2018. Patients treated with I-125 plaque radiotherapy and prophylactic intravitreal bevacizumab injections (1.25 mg/0.05 mL) at the time of plaque removal and again at 4-month intervals for a duration of 2 years were included in the study. Patients treated by other modalities or who did not receive prophylactic intravitreal bevacizumab were excluded. Informed consent was obtained from all patients.

All patients underwent a complete eye examination, including slit lamp biomicroscopy and indirect ophthalmoscopy, by a senior ocular oncologist (CLS, AM). Colour fundus photography, fundus autofluorescence, B scan ultrasonography, fluorescein angiography, and optical coherence tomography (OCT) were performed as indicated. Plaque radiotherapy was performed using I-125, with tumour localisation and plaque placement under anaesthesia. The plaque remained in place for 4–7 days, with a target radiation dose to tumour apex of 70 Grey (Gy). Intravitreal bevacizumab injections



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To cite: Dalvin LA, Zhang Q, Hamershock RA, *et al*. *Br J Ophthalmol* 2020;**104**:697–702.

were performed using standard aseptic technique with topical application of proparacaine for anaesthesia, insertion of a sterile eyelid speculum by a gloved and masked physician, and topical application of povidone iodine. Bevacizumab was injected at a dose of 1.25 mg/0.05 mL using a 30-gauge needle with trans pars plana approach, followed by postinjection indirect ophthalmoscopy. Prophylactic sector scatter retinal laser photocoagulation to target retinal ischaemia surrounding the tumour was offered to all treated patients with posterior uveal melanoma after resolution of subretinal fluid to reduce radiation side effects, if clarity of media was adequate. Records were reviewed for patient demographics, presenting tumour features, treatment parameters and outcomes.

Demographic features included age, race, sex and past medical history. Clinical features at presentation included best corrected visual acuity, presenting symptom, tumour distance to optic nerve and foveola, largest tumour basal diameter, thickness, location, associated subretinal fluid and foveolar status by OCT. Treatment parameters included plaque shape (round, notched, deep-notched), duration of application (hours), radiation dose (Gy) and dose rate (Gy/hour) to tumour apex, tumour base, optic disc, foveola and lens. Outcomes included follow-up duration (months), best corrected visual acuity at 4, 8, 12, 24, 36 and 48 months and final follow-up, lines of visual acuity loss, final tumour basal diameter and thickness, development of OCT-evident cystoid macular oedema (CMO), radiation maculopathy, radiation retinopathy, radiation papillopathy, retinal vascular occlusion, iris neovascularisation, neovascular glaucoma, scleral necrosis, tumour recurrence and enucleation.

Summary statistics are reported for demographics, clinical tumour features and treatment parameters. Time to visual outcome was defined as time of plaque application to date of

poor visual acuity (Snellen acuity worse than 20/200) or date of final follow-up. Effects of all potential prognostic factors were modelled using Cox regression. Variables were selected based on significance and using a stepwise method. Model fit was assessed based on Akaike information criterion. Two nomograms were created using R statistical software. Internal validation was done using a 10-fold cross-validation method. The nomograms were calibrated based on the prognostic models for baseline and post-treatment. Points assigned to each variable included in the nomogram were proportional to the effect size. The accuracy of predictions was evaluated by estimating the model's calibration, and discrimination was measured by the Concordance Index (C-index). A C-index of 0.5 represents agreements by chance alone, and a C-index of 1 means perfect discrimination. All other analyses were performed with SAS V9.4.

RESULTS

There were 1131 eyes of 1131 patients diagnosed with uveal melanoma and treated with plaque radiotherapy and prophylactic intravitreal bevacizumab injections at the Ocular Oncology Service, Wills Eye Hospital, Philadelphia, Pennsylvania, USA from 7 July 2000 to 1 January 2018. Demographics are listed in online supplementary table 1. Patients presented at mean age of 60 years. Race was predominantly Caucasian (n=1081, 96%), with equal sex distribution (n=557, 49% male). Past medical history included diabetes mellitus of the insulin-dependent (n=51, 5%) or non-insulin-dependent (n=105, 9%) type and systemic hypertension (n=478, 42%).

Clinical features are listed in online supplementary table 2. Mean logarithm of the minimum angle of resolution (LogMAR) visual acuity at presentation was 0.35 (Snellen

Nomogram for Visual Acuity Outcome: Clinical Factors

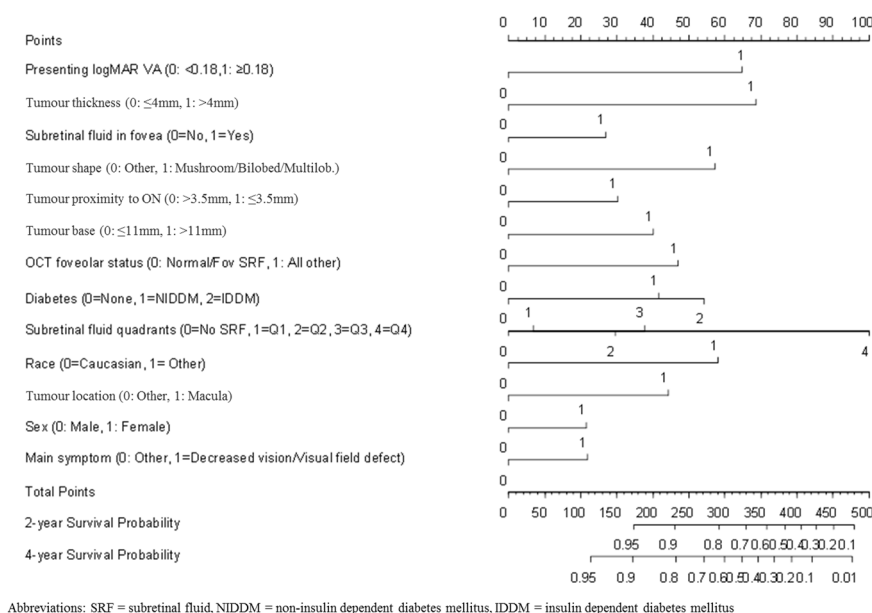


Figure 1 Nomogram for visual acuity outcome after plaque radiotherapy and prophylactic intravitreal bevacizumab for uveal melanoma using clinical risk factors. Each risk factor is assigned a point value for presence of the risk factor or 0 points for absence of the risk factor. Point values can be determined from the points scale at the top of the nomogram. The sum of point values for all present risk factors gives the total points. Total points can be read across the labelled scale and correlated to the bottom two scales, which provide the survival probability of maintaining good visual acuity (Snellen equivalent $\geq 20/200$) at 2 years and 4 years after plaque radiotherapy. The risk of poor visual acuity outcome (Snellen $< 20/200$) can be found by calculating one minus the survival probability at 2 years and 4 years. This nomogram is depicted in table format (table 1). OCT, optical coherence tomography.

equivalent 20/50). Presenting symptoms included decreased visual acuity (n=380, 34%), flashes/floaters (n=244, 22%), visual field defect (n=100, 9%), pain (n=22, 2%) or no symptoms (n=385, 34%). Mean tumour distance from optic nerve was 4.6 mm and from foveola was 4.2 mm. Mean largest tumour basal diameter was 11.5 mm and thickness was 4.8 mm. Tumour epicentre was in the macula (n=208, 18%), macula to equator (n=714, 63%), equator to ora serrata (n=168, 15%), ciliary body (n=40, 4%), or iris (n=1, <1%). Tumour configuration was dome-shaped (n=918, 81%), mushroom (n=126, 11%), plateau (n=76, 7%) or bilobed/multilobulated (n=11, 1%). Subretinal fluid was present (n=874, 77%) in 1 (n=575, 51%), 2 (n=208, 18%), 3 (n=75, 7%) or 4 (n=16, 1%) quadrants. OCT showed subfoveal subretinal fluid (n=320, 29%) or other foveolar abnormalities, including CMO, retinal atrophy, lamellar hole and epiretinal membrane (n=72, 6%).

Treatment features are listed in online supplementary table 3. Plaque shape was round (n=849, 75%), notched (n=225,

20%) or deep notched (n=57, 5%), with mean total radiation dose to tumour apex 70 Gy, tumour base 184 Gy, optic disc 38 Gy and foveola 57 Gy. Mean radiation dose rate was 72 cGy/hour to tumour apex, 185 cGy/hour to tumour base, 38 cGy/hour to optic disc and 58 cGy/hour to foveola. Prophylactic sector scatter retinal laser photocoagulation (panretinal photocoagulation, PRP) to areas of retinal ischaemia was performed a mean of 7 months (median 4, range 0–73 months) after plaque radiotherapy (n=540, 48%). Patients were treated with prophylactic intravitreal bevacizumab, with a mean of four injections.

Clinical outcomes are listed in online supplementary table 4. Mean follow-up after plaque radiotherapy was 40 months. Enucleation was required (n=15, 1%). Mean LogMAR visual acuity at final follow-up (n=1116, 99%) was 1.02 (Snellen equivalent 20/200), with poor visual acuity worse than 20/200 (n=369, 33%). Radiation side effects included iris neovascularisation (n=31, 3%), neovascular glaucoma (n=32, 3%), scleral

Table 1 Nomogram for visual acuity outcome after plaque radiotherapy and prophylactic intravitreal bevacizumab for uveal melanoma in 1131 patients: clinical risk factors at initial evaluation nomogram

Risk factor	Points	HR (95% CI)	P values
Presenting LogMAR visual acuity 0.18 or worse (Snellen 20/30 or worse)	65	2.16 (1.69 to 2.75)	<0.001
Tumour thickness >4 mm	69	2.29 (1.71 to 3.07)	<0.001
Subretinal fluid present under fovea	27	1.37 (1.05 to 1.78)	0.02
Tumour shape mushroom, bilobed or multilobulated	57	1.96 (1.50 to 2.56)	<0.001
Tumour proximity to optic nerve ≤3.5 mm	30	1.41 (1.10 to 1.79)	0.01
Tumour largest basal diameter >11 mm	40	1.60 (1.21 to 2.13)	0.001
OCT foveolar status abnormal (excluding subretinal fluid)	47	1.70 (1.21 to 2.40)	0.002
Systemic diabetes mellitus			
Non-insulin-dependent	41	1.60 (1.16 to 2.21)	0.004
Insulin-dependent	54	1.86 (1.19 to 2.93)	0.007
Subretinal fluid (quadrants involved)			
1	7	1.10 (0.79 to 1.54)	0.57
2	29	1.45 (1.02 to 2.07)	0.04
3	38	1.54 (0.98 to 2.42)	0.06
4	100	2.88 (1.45 to 5.72)	0.003
Race non-Caucasian	58		
African-American		1.06 (0.26 to 4.38)	0.93
Hispanic		2.20 (1.23 to 3.94)	0.008
Other		0.00 (0.00 to 26.00 ¹⁷⁶)	0.96
Tumour location in macula	44	1.62 (1.18 to 2.21)	0.003
Sex female	21	1.28 (1.04 to 1.58)	0.02
Main symptom of decreased vision or visual field defect	22	1.30 (1.02 to 1.64)	0.03

LogMAR, logarithm of the minimum angle of resolution.

Total points (%*)	Poor visual acuity risk 2 years/4 years (%)		Total point range	Poor visual acuity risk range 2 years/4 years (%)	
40 (30%)	11	24	≤40	0–11	0–24
69 (40%)	16	33	41 to 69	12–16	25–33
91 (50%)	21	41	70 to 91	17–21	34–41
108 (60%)	25	48	92 to 108	22–25	42–48
131 (70%)	32	59	109 to 131	26–32	49–59
173 (80%)	48	78	132 to 173	33–48	60–78
200 (90%)	60	88	174 to 200	49–60	79–88
304 (100%)	97	>99	201 to 304	61–97	89–>99
			>304	>97	>99

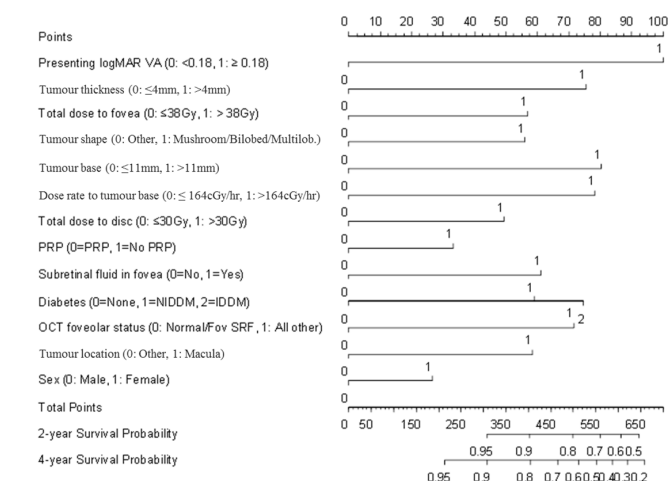
Poor visual acuity is defined as Snellen acuity worse than 20/200.

Bold values indicate significant p value.

*Denotes percentage of study patients with ≤ the listed number of points.

OCT, optical coherence tomography.

Nomogram for Visual Acuity Outcome: Clinical and Treatment Factors



Abbreviations: SRF = subretinal fluid, NIDDM = non-insulin dependent diabetes mellitus, IDDM = insulin dependent diabetes mellitus

Figure 2 Nomogram for visual acuity outcome after plaque radiotherapy and prophylactic intravitreal bevacizumab for uveal melanoma using clinical and treatment risk factors. Each risk factor is assigned a point value for presence of the risk factor or 0 points for absence of the risk factor. Point values can be determined from the points scale at the top of the nomogram. The sum of point values for all present risk factors gives the total points. Total points can be read across the labelled scale and correlated to the bottom two scales, which provide the survival probability of maintaining good visual acuity (Snellen equivalent $\geq 20/200$) at 2 years and 4 years after plaque radiotherapy. The risk of poor visual acuity outcome (Snellen $< 20/200$) can be found by calculating one minus the survival probability at 2 years and 4 years. This nomogram is depicted in table format (table 2). OCT, optical coherence tomography.

necrosis (n=12, 1%), cataract (n=302, 27%), CMO (n=413, 37%), radiation maculopathy (n=431, 38%), non-proliferative radiation retinopathy (n=269, 24%), proliferative radiation retinopathy (n=7, 1%), retinal vascular occlusion (n=325, 29%) and radiation papillopathy (n=197, 17%). Treatment for radiation side effects included intravitreal anti-VEGF (n=278, 25%) and intravitreal triamcinolone (n=156, 14%). Intravitreal triamcinolone was not used as prophylaxis in this series.

A nomogram was developed using pretreatment initial clinical factors to predict the risk of poor visual acuity outcome (Snellen worse than 20/200) at 2 years and 4 years (figure 1). Point values and HR for each risk factor are listed in table 1.

The six most important risk factors (points for nomogram) in the clinical nomogram included subretinal fluid involving four quadrants (100), tumour thickness > 4 mm (69), presenting LogMAR visual acuity 0.18 or worse (Snellen equivalent 20/30 or worse) (65), non-Caucasian race (58), tumour shape mushroom, bilobed, or multilobulated (57), and insulin-dependent diabetes mellitus (54). Less important risk factors included abnormal foveolar status by OCT other than subretinal fluid (47), tumour location in the macula (44), non-insulin-dependent diabetes mellitus (41), tumour largest basal diameter > 11 mm (40), tumour proximity to optic nerve ≤ 3.5 mm (30), subretinal fluid present under the fovea (27), main symptom of decreased vision or visual field defect (22), female sex (21), and subretinal fluid in 3 (38), 2 (29) or 1 (7) quadrants. The risk of poor visual acuity outcome at 2 years and 4 years increased with greater total number of points at 11% and

24% with 40 points, 25% and 48% with 108 points, 48% and 78% with 173 points, and 97% and $> 99\%$ with 304 points.

A second nomogram was developed using treatment factors (radiation parameters) (figure 2). Point values and HRs for each risk factor are listed in table 2.

The six most important risk factors (points for nomogram) in the treatment nomogram included presenting LogMAR visual acuity 0.18 or worse (Snellen equivalent 20/30 or worse) (100), tumour largest basal diameter > 11 mm (80), radiation dose rate to tumour base ≥ 164 cGy/hour (78), tumour thickness > 4 mm (76), insulin-dependent diabetes mellitus (75) and abnormal foveolar status by OCT other than subretinal fluid (72). Less important risk factors included subretinal fluid present under the fovea (61), tumour location in the macula (59), non-insulin-dependent diabetes mellitus (59), total radiation dose to foveola > 38 Gy (57), tumour shape mushroom, bilobed or multilobulated (56), total radiation dose to optic disc > 30 Gy (49), PRP not done after plaque treatment (33) and female sex (26). The risk of poor visual acuity outcome at 2 years and 4 years increased with greater total number of points at 6% and 14% with 56 points, 15% and 33% with 176 points, 33% and 61% with 287 points, and 88% and 99% with 496 points. The C-index for both nomograms was 0.80 (based on 10-fold cross-validation), and the calibrations plots are shown in online supplementary figures 1 and 2, which confirmed that the predicted and observed probabilities had good agreement. The nomograms are available online at <https://fighiteycancer.com/nomograms/>

DISCUSSION

Plaque radiotherapy is an effective treatment modality for small, medium and some large uveal melanomas.^{1–3} However, radiation side effects, including CMO, maculopathy, retinopathy and papillopathy, can lead to permanently reduced visual acuity in more than half of treated patients after long-term follow-up.^{1–3} Shields *et al*, in a series of 1106 patients, found factors associated with poor visual acuity outcome, including patient age ≥ 60 years, tumour location (superior, temporal, inferotemporal or superonasal fundus), tumour basal diameter ≥ 10 mm, tumour thickness > 8 mm, mushroom tumour shape, posterior pole tumour location (< 5 mm to optic disc or foveola), tumour overhang of optic disc by $\geq 50\%$, anterior tumour margin behind equator, subretinal fluid, tumour recurrence, and several radiation factors including radioactive isotope (iridium) and plaque design (notched), radiation dose to tumour apex (≥ 9000 cGy) and base ($\geq 33\ 300$ cGy), and radiation dose rate to optic disc.¹ Despite identification of these risk factors, the importance of each risk factor was not thoroughly elucidated and prediction of visual prognosis for the individual patient remained challenging. Herein, we developed two nomograms for prediction of visual acuity outcome following plaque radiotherapy for uveal melanoma using (1) Clinical risk factors. (2) Treatment risk factors.

Prior studies have developed nomograms for prediction of visual acuity outcome following plaque radiotherapy. The first nomogram by Khan *et al* found risk factors for vision loss of increased age (HR 1.01, $p=0.05$), increased tumour height (HR 1.35, $p<0.001$) and greater total radiation dose to fovea (HR 1.01, $p<0.001$).⁴ The nomogram was developed using three factors, including male sex, tumour thickness and radiation dose to the macula, to predict poor visual acuity outcome worse than 20/200 at 3 years, with thickness as the most important predictive factor.⁴ That nomogram was developed using only 189 patients that were treated with various isotopes such as

Table 2 Nomogram for visual acuity outcome after plaque radiotherapy and prophylactic intravitreal bevacizumab for uveal melanoma in 1131 patients: clinical and treatment risk factors for nomogram

Risk factor	Points	HR (95% CI)	P values
Presenting LogMAR visual acuity 0.18 or worse (Snellen 20/30 or worse)	100	2.12 (1.68 to 2.69)	<0.001
Tumour thickness >4 mm	76	1.74 (1.24 to 2.44)	0.001
Total radiation dose to foveola >38 Gy	57	1.55 (1.18 to 2.03)	0.002
Tumour shape mushroom, bilobed or multilobulated	56	1.51 (1.16 to 1.98)	0.002
Tumour largest basal diameter >11 mm	80	1.25 (0.98 to 1.58)	0.07
Radiation dose rate to tumour base >164 cGy/hour	78	1.80 (1.35 to 2.40)	<0.001
Total radiation dose to optic disc >30 Gy	49	1.44 (1.12 to 1.85)	0.004
Panretinal photocoagulation not done after treatment	33	1.27 (1.03 to 1.56)	0.02
Subretinal fluid present under fovea	61	1.56 (1.23 to 1.99)	<0.001
Systemic diabetes mellitus			
Non-insulin-dependent	59	1.54 (1.11 to 2.13)	0.01
Insulin-dependent	75	1.74 (1.11 to 2.72)	0.02
OCT foveolar status abnormal (excluding subretinal fluid)	72	1.69 (1.21 to 2.38)	0.002
Tumour location in macula	59	1.51 (1.10 to 2.06)	0.01
Sex female	26	1.22 (0.99 to 1.50)	0.06

LogMAR, logarithm of the minimum angle of resolution.

Total points (%*)	Poor visual acuity risk 2 years/4 years (%)		Total point range	Poor visual acuity risk range 2 years/4 years (%)	
56 (10%)	6	14	≤56	0–6	0–14
80 (20%)	8	17	57 to 80	7–8	15–17
135 (30%)	11	25	81 to 135	9–11	18–25
176 (40%)	15	33	136 to 176	12–15	26–33
206 (50%)	19	39	177 to 206	16–19	34–39
234 (60%)	23	47	207 to 234	20–23	40–47
287 (70%)	33	61	235 to 287	24–33	48–61
340 (80%)	46	76	288 to 340	34–46	62–76
396 (90%)	61	89	341 to 396	47–61	77–89
496 (100%)	88	99	397 to 496	62–88	90–99
			>496	>88	>99

Poor visual acuity is defined as Snellen acuity worse than 20/200.

Bold values indicate significant *p* value.

*Denotes percentage of study patients with ≤ the listed number of points.

OCT, optical coherence tomography.

Ru-106 or I-125 and was developed prior to the implementation of prophylactic intravitreal anti-VEGF injections.⁴ A second nomogram by Tagliaferri *et al* based on 197 patients with uveal melanoma treated with Ru-106 brachytherapy, was designed to predict radiation maculopathy at 3 years.⁵ Risk factors for radiation maculopathy included tumour distance to foveola (HR 0.83, *p*<0.01), diabetes mellitus (HR 2.92, *p*<0.01) and tumour volume (HR 21.6, *p*=0.02), with shorter tumour distance to foveola as the most important risk factor.⁵ This series studied patients treated with Ru-106 plaque radiotherapy and without prophylactic intravitreal anti-VEGF.⁵ Thus, there was a need for a new model to predict visual acuity outcome in patients treated with I-125 plaque radiotherapy in the intravitreal anti-VEGF era. Our models were developed using 1131 patients treated with I-125 plaque radiotherapy and prophylactic intravitreal bevacizumab injections given at 4-month intervals for a total duration of 2 years. Although we found risk factors similar to those identified in prior studies^{1,4,5} our larger cohort allowed us to develop a more accurate model for individualised prediction of visual acuity outcome.

Due to the guarded visual prognosis of postradiation macular oedema, even with treatment,^{6–13} our group has

tried to identify methods to prevent development of macular oedema after plaque radiotherapy of uveal melanoma. Since 2008, our practice has offered prophylactic treatment with intravitreal bevacizumab at 4-month intervals at the time of plaque removal and for a total duration of 2 years following plaque radiotherapy.³ In our initial 2-year follow-up study of 292 patients, we found that bevacizumab-treated patients were less likely to have poor visual acuity worse than 5/200 when compared with controls treated without bevacizumab (15% vs 28%, *p*=0.004).³ Other benefits of prophylactic bevacizumab at 2 years follow-up included less OCT-evident CMO (26% vs 40%, *p*=0.004), less clinically evident radiation maculopathy (16% vs 31%, *p*=0.001) and less moderate vision loss (33% vs 57%, *p*<0.001).³ We recently reported 4-year visual acuity outcomes in a series of 1131 patients and found that patients treated with bevacizumab every 4 months for a duration of 2 years after radiation were less likely to have visual acuity of 20/200 or worse on final follow-up compared with controls (36% vs 52%, *p*=0.002).¹⁴ Median visual acuity 48 months after plaque radiotherapy for patients managed with bevacizumab versus controls was 20/70 versus count fingers (*p*<0.001).¹⁴ Thus, we have found prophylactic intravitreal

bevacizumab to be an important factor in improving long-term visual acuity outcome in our patients treated with plaque radiotherapy, even 4 years after treatment.¹⁴ In the current study, we included only patients managed with this updated practice pattern in order to best prognosticate the long-term risk of poor visual acuity for our patients.

Nomograms can be important tools in medicine, especially in oncology.¹⁵ A well-designed nomogram allows a clinician to generate the numerical probability of a clinically significant event for an individual patient by inputting a series of demographic, clinical, genetic and/or treatment parameters unique to that patient.¹⁵ In our study, we developed two nomograms. The first nomogram was developed using pretreatment clinical factors at initial evaluation, while the second nomogram combined clinical findings and radiation parameters. Using the nomograms, each risk factor is assigned 0 points if (absent) not associated with poor visual acuity outcome or >0 points if (present) associated with poor visual acuity outcome (figures 1 and 2). The number of points assigned to each risk factor is read off a scale at the top of the nomogram. The point values for each risk factor are added to determine the total points for an individual patient. Total points are found on a labelled scale at the bottom of the nomogram and are correlated to survival probability scales for 2-year and 4-year visual acuity outcome. The survival probability indicates the likelihood of maintaining good visual acuity (Snellen equivalent $\geq 20/200$) at 2 years and 4 years after plaque radiotherapy. The risk of poor visual acuity outcome (Snellen $<20/200$) can be found by calculating one minus the survival probability at 2 years and 4 years.

Study limitations include a retrospective design and variable adherence to the prophylactic bevacizumab injection regimen. Some patients received additional treatment for radiation side effects, including monthly intravitreal anti-VEGF and intravitreal triamcinolone injections, which could have affected visual acuity outcome. Nevertheless, treatments for radiation side effects were not significant predictors of visual acuity outcome in this study. Study strengths include a large cohort with a single planned treatment regimen (I-125 and prophylactic bevacizumab) allowing for a more uniform study population. We included more risk factors than prior nomograms, which might allow for more individualised vision prognostication. However, caution should be used when applying these nomograms in different practice settings. External validation of the nomograms will need to be done at a future date.

In summary, we have developed two nomograms to predict the risk of poor visual acuity outcome (Snellen acuity worse than 20/200), one using clinical risk factors and a second using clinical and treatment factors, including radiation dosage parameters. The nomograms are able to provide individualised visual acuity prognostication at 2 years and 4 years after treatment for patients with uveal melanoma managed with plaque radiotherapy and prophylactic intravitreal bevacizumab. The most important risk factors in the clinical nomogram included subretinal fluid involving four quadrants, tumour thickness >4 mm, presenting visual acuity 20/30 or worse, non-Caucasian race, tumour shape mushroom, bilobed or multilobulated, and insulin-dependent diabetes mellitus, and in the treatment nomogram included presenting visual acuity 20/30 or worse, tumour largest basal diameter >11 mm, radiation dose rate to tumour base ≥ 164 cGy/hour, tumour thickness >4 mm, insulin-dependent diabetes

mellitus and abnormal foveolar status by OCT at presentation. Future external validation is required to determine the utility of these nomograms in other practice settings.

Contributors The authors have contributed by conception of the study (LAD, CLS) with supervision (AM, CLS) and critical revision of the manuscript (QZ, RAH, MC, MDY, AM, CLS), data collection (LAD, MC, MDY), drafting of the manuscript (LAD), and statistical analysis (QZ, RAH). CLS has had full access to all the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.

Funding This study was partly supported by the Eye Tumor Research Foundation, Philadelphia, Pennsylvania (CLS). The funders had no role in the design and conduct of the study, in the collection, analysis and interpretation of the data, and in the preparation, review or approval of the manuscript.

Competing interests None declared.

Patient consent for publication Not required.

Ethics approval Institutional Review Board/Ethics Committee approval was obtained from Wills Eye Hospital. The study adhered to the tenets of the Declaration of Helsinki and the Health Insurance Portability and Accountability Act.

Provenance and peer review Not commissioned; externally peer reviewed.

Data availability statement All data relevant to the study are included in the article or uploaded as supplementary information.

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